



Adaptive Online Continual Learning for In-Situ Quality Prediction in Manufacturing Processes

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Manufacturing processes undergo continuous changes to meet various requirements, such as process/product changes and variations in tool/workpiece conditions, leading to mixed, heterogenous, or anomalous data. As a result, a quality prediction model trained from previous data may not perform well when new tasks emerge. To achieve in-time and accurate product quality prediction, it is crucial to develop a predictive method that adapts to variations in the manufacturing system, capable of learning from new tasks without forgetting previous ones and detecting unknown tasks. This study proposes a deep learning method integrated with continual learning for in-situ quality prediction that is capable of learning from new tasks without forgetting previous ones. To demonstrate this idea, deep convolutional neural networks (CNNs) are designed to analyze in-process sensor data, which consist of shared layers to capture the common underlying features across all tasks, and task-specific layers that capture specific characteristics of each individual task. To identify the task to which the incoming product belongs, a task prediction approach based on task relevancy using filter subspace distance is proposed. When new data come in, the model first identifies the task, followed by predicting the quality of the current product. The proposed method is demonstrated in two case studies, including quality prediction of the workpiece using acoustic emissions during the laser-induced plasma micromachining process and quality prediction of the product through thermal images during the hot stamping process. [DOI: 10.1115/1.4066799]

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1 Introduction

Accurate in-situ quality prediction ensures that products meet required specifications and quality standards, thereby enhancing customer satisfaction, reducing waste, and increasing profitability. Recently, advancements in sensor technology, such as the use of acoustic emission (AE) signals in laser machining processes, pyrometers, and the integration of machine learning models, have presented unique opportunities for developing innovative methods for in-situ quality prediction in manufacturing. Real-time data in the manufacturing process, such as temperature and vibration, are collected by sensors in the form of images or signals. Machine learning and deep learning algorithms are capable of analyzing such complex data. For instance, support vector machine regression models are used for analyzing acoustic emission signals, while deep convolutional neural networks (CNNs) are employed in processing thermal images. These methods have been widely demonstrated in various manufacturing processes, including welding

[1,2], machining [3,4], and additive processes [5,6]. Accordingly, accurate and timely predictions can be made for product quality, facilitating the identification of potential defects and enabling informed decision-making in the manufacturing process.

Advanced technologies have shown immense potential in leveraging rich sensing data for in-situ monitoring. However, the evolution of the manufacturing paradigm from mass production to mass individualization brings about challenges of adapting to high variability. There are higher product variety and lower volume per product variant due to customized features and shorter product life cycles [7]. Previously, manufacturing concentrated on single tasks, but the current demand requires handling a variety of tasks. In manufacturing, “one task” refers to the same product design or process, where features and data distribution remain consistent [8]. Manufacturing processes are constantly changing to meet various requirements, such as modifications to the process or product [9], alterations in tool [10] or machine conditions [11], and changes in raw materials [12]. These changes can lead to variations in output products, and data are mixed and heterogeneous. This renders previous criteria for quality prediction less suitable and can bring about “new tasks” in the manufacturing process, resulting in the need for new models to be developed. Moreover, tools and machines can be

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maintained, repaired, or replaced, and the same product or process design may be in demand periodically or irregularly, leading to the reemergence of a task. Additionally, changes in machine condition, product design, or process design may not be known ahead of time, and thus, unknown tasks emerge during the manufacturing process. These tasks often involve anomalous data with unpredictable distributions and characteristics, making it challenging to develop a suitable model.

With the emergence of diverse tasks, conventional in-situ quality prediction methods fall short of addressing the dynamic changes that occur during the manufacturing process. One potential solution to this problem is to train individual models for each task. However, this approach is not feasible due to its high computational and time costs. Furthermore, some tasks are short-lived, resulting in insufficient historical data to train an effective model. To tackle this challenge, multitask learning is a promising technique that can jointly train a model for multiple datasets while exploiting commonalities and differences across tasks [13,14]. However, multitask learning has limitations, as the model training relies on the data collected from all the tasks. This may bring about challenges to data storage in manufacturing processes where a large number of data are sequentially collected from different tasks. Transfer learning is a promising technique that can transfer a pretrained model to a similar dataset and require less data and computational time [9,12]. However, transfer learning has limitations, as the model may forget previous tasks when learning new ones. This can lead to relatively low performance when a task reemerges. To overcome this issue, we propose a continual learning approach that prevents catastrophic forgetting of previous tasks. Furthermore, to handle the emergence of unknown tasks during the manufacturing process, the model is improved to support online continual learning. This enables the model to adapt to new tasks in real time, without the need for extensive retraining.

The objective of this article is to develop a product quality prediction method tailored to the manufacturing system's variability. It should learn from new tasks while retaining knowledge of past ones and identify unknown tasks to prevent the catastrophic forgetting of previously learned information. The proposed method includes a deep CNN model that can process acoustic signals and images for manufacturing quality prediction, a novel online task identification algorithm to deal with unknown tasks, and an online continual learning algorithm that can accommodate reemerge tasks and unknown tasks. The method is implemented in two case studies. Specifically, one focuses on the quality prediction of the workpiece using acoustic emissions during the laser-induced plasma micromachining process, utilizing a deep CNN, and the other one investigates the quality prediction of the product through thermal images during the hot stamping process. The proposed method shows its effectiveness in task identification and quality prediction.

The remainder of this article is structured as follows: in Sec. 2, a comprehensive literature review is presented. Section 2.1 summarizes existing studies on in-situ predictive quality in manufacturing, while Sec. 2.2 reviews current progress in continual learning and highlights its limitations. Furthermore, Sec. 2.3 examines the application of continual learning in predictive quality in manufacturing. In Sec. 3, we introduce our proposed online continual learning approach, which includes a task identification method and predictive models. Section 4 illustrates the efficacy of our method by analyzing two case studies in different manufacturing domains. Finally, Sec. 5 concludes the article and provides suggestions for future research extensions.

2 Literature Review

2.1 In-Situ Predictive Quality in Manufacturing. In-situ predictive quality is crucial for ensuring that each product meets the required specifications and quality standards, maintaining customer satisfaction, reducing waste, and increasing profits. This is

achieved through the use of advanced sensing technologies that monitor the production process, such as acoustic emissions or pyrometers, and the application of deep learning-based models that can effectively analyze the data collected.

By leveraging sensor data such as signals or images, recent studies have explored the potential of in-situ predictive quality for manufacturing processes. Machine learning or deep learning models have been developed to analyze the data and make accurate predictions. Wu et al. [1] developed a machine learning approach that integrates CNN and random forest classifier to predict the keyhole inside the weld pool during the variable polarity plasma arc welding process, leveraging the welding images. Taherkhani et al. [15] proposed a self-organizing map-based unsupervised learning algorithm for in-situ defect-detection in laser powder-bed fusion utilizing the captured light intensity signal. Khanzadeh et al. [16] developed an in-situ signal processing method for characterizing the thermo-physical dynamics of directed energy deposition melt pool signals, which can be subsequently used for quality prediction of the products. Chen and Guo [5] developed a deep learning-based method integrating deep convolutional generative and adversarial models and CNN for in-situ quality prediction by detecting porosity during the additive manufacturing process, making use of the thermal images collected by the pyrometer. However, existing in-situ quality prediction methods are under the assumption that no changes occur during the manufacturing process. Indeed, changes in the manufacturing process can arise due to various reasons. Alterations in the design or specifications of the product can necessitate adjustments to the manufacturing process. Furthermore, variations in the raw materials employed in the manufacturing process can also bring about changes. The manufacturing equipment's performance and the product's quality can both be affected by factors such as temperature, humidity, and pressure. Moreover, deviations in tool or machine conditions can influence the quality of the end product. Those lead to the model's inability to perform well when new tasks emerge. To mitigate this challenge, there is a pressing need for methods that can effectively adapt to the variations in the manufacturing system.

Targeting on improving models' adaptation to systems variation, some studies have investigated in-situ predictive quality using transfer learning. The fundamental principle of transfer learning is to transfer a pretrained deep learning model to a similar dataset. During training, the parameters of most layers are fixed, and only the last layers are fine-tuned with a small learning rate. Pandiyan et al. [12] made use of AE signals with VGG-16 and ResNet-18 models for in-situ monitoring and quality prediction in laser powder bed fusion. Chen et al. developed a transferrable method for in-situ process monitoring and quality prediction in laser-induced plasma micromachining (LIPMM). They investigated the feasibility of transfer learning and compared the performance of various transfer learning models: different input features, different CNN structures, and the same structure with different fine-tuned layers [3]. The results demonstrated that transfer learning can be generalized to a variety of tasks, requiring less data and less computational time.

Although transfer learning models can deal with new tasks with less data requirement and computational cost, they are likely to forget previous tasks when learning from new tasks. In a manufacturing process, an old task may reemerge due to various reasons such as renewal of raw materials, tool or machine conditions, or other process parameters. If the transfer learning model forgets previous tasks, it will require retraining on the entire dataset to include the forgotten tasks, which can be time-consuming and computationally expensive. Therefore, we need to develop a method that allows the model to adapt to new tasks without forgetting the previous ones.

2.2 Continual Learning Models. Aiming at avoiding catastrophic forgetting of previous tasks, continual learning has proven to be a valuable approach in many datasets. Continual learning is a machine learning paradigm that focuses on the ability of a

model to learn continuously from new data while avoiding catastrophic forgetting of previously learned knowledge, where the models can be divided into three categories: replay methods, regularization-based methods, and parameter isolation methods [17].

Replay methods involve storing samples in their raw format or generating pseudo-samples using a generative model. These samples from previous tasks are then replayed during the learning process of a new task to alleviate forgetting. They are either reused as model inputs for rehearsal or are used to constrain the optimization of the new task loss to prevent interference with previous tasks. The model does not require the total number of tasks so that it can learn from the newly added task, but it requires a relatively long training time. *Regularization-based methods* have been developed to address privacy concerns and reduce memory requirements by eliminating the need to store raw input data. These methods can be further divided into two categories: prior-focused and data-focused approaches. Prior-focused methods work by penalizing changes to important parameters during the training of new tasks. One such method is elastic weight consolidation (EWC), proposed by Kirkpatrick et al. [18]. Data-focused methods, on the other hand, rely on knowledge distillation, where a model trained on a previous task acts as a teacher to guide the learning of a new task. A key example of this approach is learning without forgetting (LwF), proposed by Li and Hoiem [19], which utilizes knowledge distillation to transfer knowledge from the teacher model to the student model as it learns from new data. Unlike replay methods, regularization-based methods do not need to store historical data, which saves lots of storage costs. However, it requires similarities among all tasks to guarantee the prediction accuracy of all tasks. In addition, from task A to task B, we need to consider all parameters in the network, thus leading to a relatively long training time. *Parameter isolation methods* dedicate different model parameters to each task to prevent any possible forgetting. When no constraints apply to architecture size, one can grow new branches for new tasks, while freezing previous task parameters. Representative models were Progressive Neural Networks [20], PackNet [21], and Supermasks in Superposition model [22]. This method has a limitation in terms of accommodating a limited number of tasks. As the number of tasks increases, models may lose effectiveness in preserving knowledge from all the previous tasks.

Based on the methods introduced above, some variations have been made to improve the performance. Table 1 summarizes recent work in continual learning. The “need task label?” column indicates that the model relies on information regarding the specific task to which the incoming data belongs. On the other hand, the “task incremental?” column signifies the model’s capability to identify and handle previously unseen tasks. Some works tried to

improve the prediction accuracy by developing innovative loss functions [23]. Some works reduced the memory cost by simplifying the stored data [24–27]. However, a comprehensive review of the progress made on continual learning reveals that a significant portion of existing methods for continual learning require task information, which mandates the predefinition of tasks (i.e., prior knowledge of the task label) before training the predictive model.

Although the method proposed by Aljundi et al. [30] does not require task labels and can be applied incrementally, it has limitations that our approach aims to address. Their method mitigates forgetting by penalizing changes to critical parameters during training, thereby minimizing the impact of new tasks on previously learned knowledge. However, as more tasks are added incrementally, the accumulated parameter adjustments can cause the model’s parameters to drift from their optimal values for earlier tasks. This drift becomes particularly problematic when the new tasks differ significantly from the initial ones, as the penalization strategy may no longer effectively preserve the essential knowledge required for those earlier tasks. Consequently, if previously learned tasks reemerge after the model has been fine-tuned for several new tasks, the prediction performance on these earlier tasks is likely to degrade. This decline in performance could limit the model’s adaptability and overall effectiveness. Therefore, there is a need to overcome these challenges by developing a more flexible and robust solution for handling diverse and evolving tasks.

2.3 Continual Learning in Manufacturing. Inspired by the success of continual learning in the domain of pattern recognition and image processing, some investigations have been made on the application of continual learning for anomaly detection and quality prediction in manufacturing, making use of rich sensor data during the manufacturing process. Maschler et al. applied continual learning for anomaly detection in discrete manufacturing so that the model can adapt to changes in different products/production processes [33]. They compared different regularization approaches (different loss functions) in continual learning, including EWC, LwF, and Synaptic Intelligence. Tercan et al. [9] proposed a memory-aware synapses method for quality prediction in injection molding.

The model proposed by the researchers has a limitation in that it requires prior knowledge of task labels for making predictions, and it is unable to recognize new tasks. This can be problematic in manufacturing processes where certain changes such as machine degradation or tool wear may not be directly observable, necessitating additional costs to supervise machine or tool condition. Additionally, new target products or machine conditions can emerge during production, rendering previously trained models unsuitable

Table 1 Summary of recent works in continual learning

Article	Approach	Stored data	Contribution/innovation	Need task label?	Task incremental?
[23]	Replay method	Raw data	An innovative loss function	Yes	Yes
[24]		Episodic memories	Reduced memory cost	Yes	No
[28]		Raw data	Can be used for stream data	Yes	No
[25]		Average episodic memories	Reduced memory cost	Yes	No
[26]	Data-focused method	Logits	Reduced memory cost	Yes	Yes
[29]		No	Estimate the importance of all neural networks using Bayesian estimation	Yes	Yes
[30]		No	Estimate the importance of all neural networks using Memory Aware Synapses	No	Yes
[31]	Prior-focused method	Logits	Unique knowledge distills parameters for each task	Yes	Yes
[32]		Encoded data	Introduced a memory recovery paradigm in which we query the network to synthesize past exemplars whenever a new task emerges	Yes	Yes
[27]		Atom memory	Reduced memory cost	Yes	Yes

for predicting the quality of new tasks. As such, an online prediction algorithm is required to address this issue. Hence, there is a need to develop a continual learning method that incorporates task prediction and allows for online prediction in quality prediction for manufacturing processes.

2.4 Summary of Contributions. A detailed review of the collected literature indicates that only a few studies have discussed the continual learning approaches that can address unknown and unseen tasks in the manufacturing process. On the other hand, the literature dealing with online continual learning in manufacturing is fairly broad. Several variants of continual learning methods have been studied, including replay methods, regularization-based methods, and parameter isolation methods. Considering the existing trends in in-situ predictive quality of the manufacturing process and potential benefits that can be offered by online continual learning, our goal is to develop a method that can adapt to variations in the manufacturing system, capable of learning from new tasks without forgetting previous ones and detecting unknown tasks. The method involves a deep learning model integrated with continual learning for in-situ quality prediction. This study aims to provide the following contributions to the state-of-the-art:

- Deep CNN models to process acoustic signals and images for manufacturing quality prediction. The model consists of shared layers to capture the common underlying features across all tasks and task-specific layers that capture specific characteristics of each individual task.

- An online task identification method based on task relevance to deal with unknown tasks. When new data come in, the model first identifies the task, followed by predicting the quality of the current product.
- An online continual learning algorithm that can accommodate reemerge tasks and unseen tasks.

3 Methods

The proposed method consists of three main components. First, deep CNN models are employed to analyze sensor information, such as AE emission and thermal images. These CNN models are tailored to the specific dataset and will be described in detail in Sec. 4. Second, to identify the task to which the incoming product belongs, a task prediction approach based on task relevancy using filter subspace distance is proposed [9]. This allows for the recognition of unknown tasks. Third, a continual learning approach is introduced to enable the model to learn from new tasks without forgetting previous ones. The working mechanism of the proposed method is illustrated in Fig. 1. Initially, historical data from task 1 are used to train both the shared parameters and the task-specific parameters for task 1. When task 2 data are introduced, the shared parameters are fine-tuned to integrate the new information from task 2 while preserving knowledge from task 1. Simultaneously, task-specific parameters for task 2 are developed to address the unique aspects of the new task. When new data arrive, the proposed task prediction method is employed to identify its associated task. For new tasks, predictions are made using the

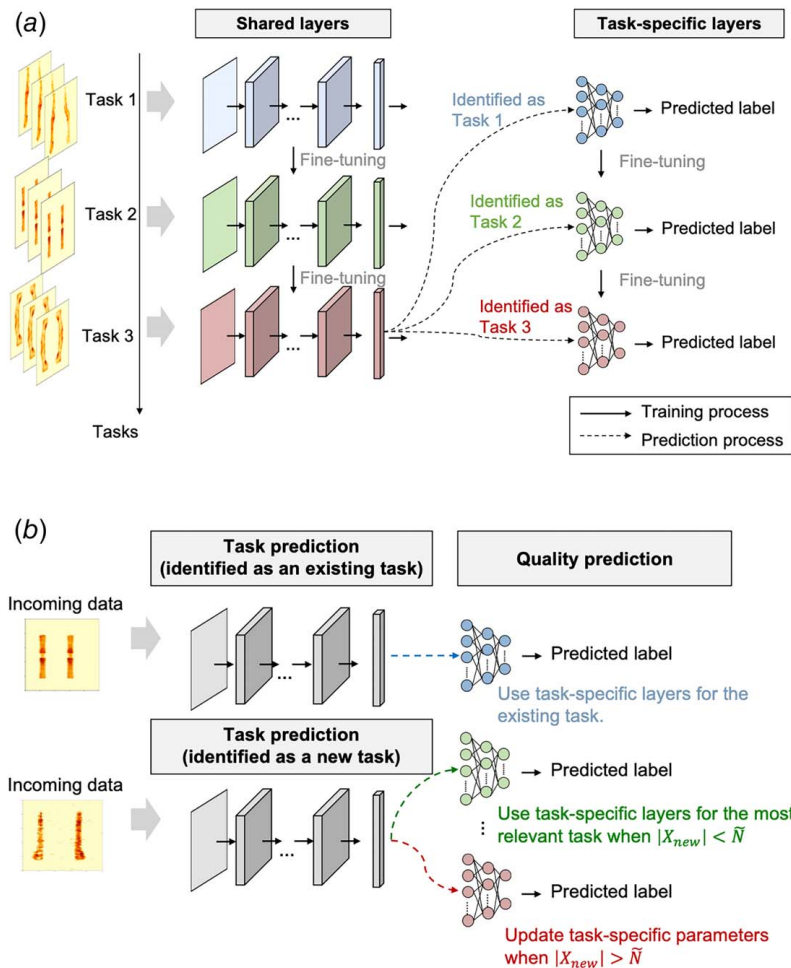


Fig. 1 Working mechanism of the proposed method: (a) training procedure and (b) online prediction procedure

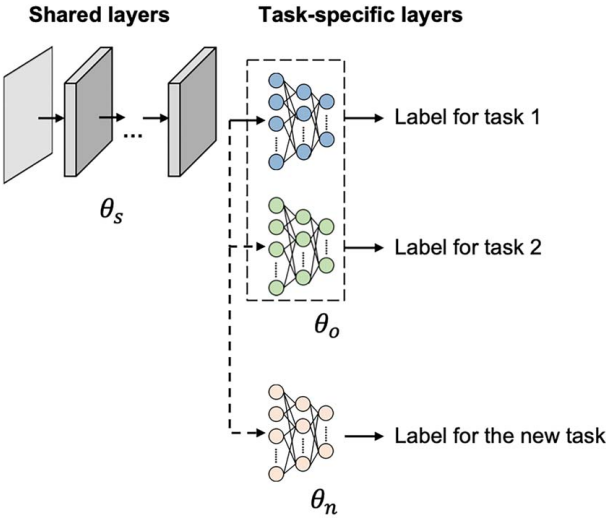


Fig. 2 An example of the proposed continual learning model structure

task-specific layers of the most relevant task. As the number of data $|X_{\text{new}}|$ in such a task reaches a certain threshold $\tilde{N}$, the shared parameters are fine-tuned using all accumulated data, and the task-specific layers are updated for the new task.

3.1 The Proposed Continual Learning Model. The proposed continual learning model is supposed to identify unseen tasks and make prediction for both existing and new tasks. Figure 2 illustrates an example of the model structure. Shared layers and task-specific layers are designed to address the challenge of learning multiple tasks sequentially without forgetting previously learned information. We follow the guideline of Li and Hoiem [19] but design our own continual learning framework. By task-sharing layers, the network can capture common patterns and knowledge that can be transferred and utilized across multiple tasks. Task-specific layers allow the network to adapt and specialize its predictions or outputs based on the unique requirements of each task. We define θ_s as shared parameters in a CNN model, θ_o as task-specific parameters for old tasks, and θ_n as task-specific parameters for new tasks. Training shared parameters and task-specific parameters involves joint training to learn common features across tasks and fine-tuning to adapt task-specific layers. In joint training, the shared parameters are optimized to minimize the overall loss across all tasks. In fine-tuning, task-specific layers are trained separately while keeping the shared layers fixed.

The ground truth label for old tasks is denoted as y_o , and the predicted label for old tasks is denoted as $\hat{y}_o$. We define the loss function L_{old} for old tasks, represented by Eq. (1).

$$L_{\text{old}}(y_o, \hat{y}_o) = -H(y_o', \hat{y}_o') = -\sum_{i=1}^l y_o^{(i)} \log \hat{y}_o^{(i)} \quad (1)$$

where l is the number of labels and $y_o^{(i)}$ and $\hat{y}_o^{(i)}$ are the modified versions of recorded and current probabilities $y_o^{(i)}$ and $\hat{y}_o^{(i)}$, which are produced by using a “softmax” output layer that converts the logit, shown by Eq. (2). Followed by the guidance of Refs. [19,34], we use $M = 2$:

$$y_o^{(i)} = \frac{(y_o^{(i)})^{\frac{1}{M}}}{\sum_j (y_o^{(j)})^{\frac{1}{M}}}, \quad \hat{y}_o^{(i)} = \frac{(\hat{y}_o^{(i)})^{\frac{1}{M}}}{\sum_j (\hat{y}_o^{(j)})^{\frac{1}{M}}} \quad (2)$$

Algorithm 1 illustrates the proposed continual learning model. The continual learning algorithm is designed to address the

challenge of catastrophic forgetting while efficiently learning new tasks. It begins by initializing shared parameters (θ_s), which provide a baseline representation of features that are generally applicable across tasks. Task-specific parameters (θ_o) for old tasks are also initialized to retain unique features of previously learned tasks. A loss balance weight (λ_0) is set to appropriately balance the losses of old and new tasks, ensuring that the model does not focus solely on new tasks at the expense of forgetting older ones. We set $\lambda_0 = 1$ for our experiment, implying that the losses associated with both old and new tasks will be given equal weight in the combined loss function. Outputs for old tasks (Y_o) are computed using existing parameters on training data. During training, the algorithm optimizes the parameters ($\theta_s, \theta_o, \theta_n$) to minimize a combined loss that includes old task loss (L_{old}), new task loss (L_{new}), and a regularization term (R). This approach ensures the model effectively retains knowledge of old tasks, learns new ones, and prevents overfitting.

Algorithm 1. The proposed continual learning algorithm

Start with

θ_s : shared parameters

θ_o : task-specific parameters for old tasks

λ_0 : loss of balance weight

X_n, Y_n : training data and ground truth on the new task

Initialize

$Y_o \leftarrow \text{CNN}(X_n, \theta_s, \theta_o)$, compute the output of old tasks for new data

$\theta_n \leftarrow \text{RandInit}(X_n, \theta_s, \theta_o)$, randomly initialize new parameters

Train

Define $\hat{Y}_o \equiv \text{CNN}(X_n, \hat{\theta}_s, \hat{\theta}_o)$, old task output

Define $\hat{Y}_n \equiv \text{CNN}(X_n, \theta_s, \theta_n)$, new task output

$$\theta_s^*, \theta_o^*, \theta_n^* \leftarrow \underset{\hat{\theta}_s, \hat{\theta}_o, \hat{\theta}_n}{\text{argmin}} (\lambda_0 L_{\text{old}}(Y_o, \hat{Y}_o) + L_{\text{new}}(Y_n, \hat{Y}_n) + R(\hat{\theta}_s, \hat{\theta}_o, \hat{\theta}_n))$$

3.2 Online Task Prediction Algorithm. To identify the task that the incoming product belongs to, we propose an online task prediction algorithm based on task relevancy by filter subspace distance [27]. In this study, the filter refers to the task-shared layers trained by existing tasks, and the subspace refers to the features extracted by the task-shared layers. The subspace distance effectively captures the most representative information of a product. By evaluating the distance between subspaces, we can measure how well a given subspace fits a particular task. A shorter distance or higher similarity value indicates greater relevancy of the subspace to the specific task.

In this algorithm, $\mathbf{v}_{\text{new}}$ denotes the subspace of the new incoming product, while $\mathbf{v}_1^{(i)}, \mathbf{v}_2^{(i)}, \dots, \mathbf{v}_{N_i}^{(i)}$ represent subspaces of task t , where N_i is the number of products in task t . The algorithm computes subspace distances using the Euclidean distance $\|\mathbf{v}_1 - \mathbf{v}_2\|_2$, which enables us to assess similarity. The relevancy between the incoming product and task t , denoted as $D^{(t)}$, is calculated by averaging the subspace distances $(1/N_i) \sum_{k_i=1}^{N_i} \|\mathbf{v}_{\text{new}} - \mathbf{v}_{k_i}^{(i)}\|_2$. To distinguish between existing and new tasks, we define a relevancy threshold $\tilde{D}^{(t)} = \max \left\{ (1/N_i) \sum_{k_i=1}^{N_i} \|\mathbf{v}_i - \mathbf{v}_{k_i}^{(i)}\|_2, i = 1, 2, \dots, N_i \right\}$, representing the maximum within-task distance. This threshold represents the upper limit of what is considered normal variation within task t . Using the maximum average distance ensures that the threshold covers the full range of acceptable variations, allowing the model to recognize all valid instances of a task. If the relevancy between the incoming product and task t exceeds the threshold $\tilde{D}^{(t)}$ for all existing tasks, it is identified as a new task since it is irrelevant to existing tasks. Otherwise, it will be assigned to the task with the least distance. This approach ensures that the model can effectively classify incoming products based on task relevancy. By leveraging the subspace distance, the algorithm can capture nuanced differences and similarities between tasks, enabling accurate identification and assignment of products to the corresponding task.

Algorithm 2. The proposed online task prediction algorithm**Start with**

θ_s : shared parameters
 x_{new} : incoming product
 X_{new} : training dataset of the new task
 T : the number of existing tasks

Compute

$v_{\text{new}} \leftarrow \text{CNN}(x_{\text{new}}, \theta_s)$, compute the subspace based on the task-shared layers.
 $D^{(t)} \leftarrow (1/N_t) \sum_{k=1}^{N_t} \|v_{\text{new}} - v_{k_i}^{(t)}\|_2, t = 1, 2, \dots, T$, compute the task relevancy with task t .

$\tilde{D}^{(t)} \leftarrow \max \left\{ (1/N_t) \sum_{k=1}^{N_t} \|v_i - v_{k_i}^{(t)}\|_2, i = 1, 2, \dots, N_t \right\}$, compute the relevancy threshold of task t .

if $D^{(t)} > \tilde{D}^{(t)}, \forall t$

$T \leftarrow T + 1$

$l_{\text{new}}^* \leftarrow T$

$X_{\text{new}} \leftarrow x_{\text{new}}$, add the new data to the training dataset of the new task.

else

$l_{\text{new}}^* \leftarrow \argmin \{ \min \|v_{\text{new}} - v_{k_i}^{(t)}\|_2, k_i = 1, 2, \dots, N_t \}$

end $D^{(t)} < \tilde{D}^{(t)}$

3.3 Quality Prediction Mechanism. Upon task identification, the prediction of quality labels for the associated products is facilitated by task-specific layers. However, during the early stages of a newly identified task, the available data may be insufficient for effective training of the task-specific layers. To address this challenge, we propose initially using the task-specific layers from the most similar, existing task to predict the quality of the first few products of the new task. As the historical data gradually accumulate and reach a predetermined threshold, denoted as $\tilde{N}$, the task-specific layers are then trained using the expanded historical data. Algorithm 3 illustrates the algorithm of the proposed quality prediction mechanism. For any existing task labels ($t = 1, 2, \dots, T_o$), if the predicted label of the incoming product (l_{new}^*) matches an existing task, the quality label (y_{new}^*) is computed using the corresponding shared parameters (θ_s) and task-specific parameters ($\theta_o^{(t)}$). The model is expressed as $\text{CNN}(X_n, \theta_s, \theta_o^{(t)})$. However, if the incoming product is predicted/identified as a new task, the task-specific layer of the most relevant task is used for quality label prediction due to the lack of sufficient training data. The most relevant task (t') is determined by $\argmin_t \left\{ (1/N_t) \sum_{k=1}^{N_t} \|v_{\text{new}} - v_{k_i}^{(t)}\|_2, t = 1, 2, \dots, T \right\}$, which compares the average subspace distance across each task. Once the dataset reaches the threshold size $\tilde{N}$, the continual learning model is expanded with X_n and Y_{T+1} , the training dataset of the new task. The shared layer and the task-specific layer are updated by

minimizing the combined loss, which includes the old task loss (L_{old}), the new task loss (L_{new}), and a regularization term (R). The shared parameters are optimized from θ_s to θ_s^* , and the task-specific parameters are optimized from $\theta_o^{(t')}$ to θ_{T+1}^* . This strategy ensures that the model can provide reliable quality predictions, even for new tasks, by adapting to changing conditions while maintaining accuracy.

Algorithm 3. The proposed quality prediction mechanism**Start with**

θ_s : shared parameters
 $\theta_o^{(1)}, \theta_o^{(2)}, \dots, \theta_o^{(T_o)}$: task-specific parameters for each task
 $X_{\text{new}}, Y_{\text{new}}$: training dataset of the new task
 x_{new} : incoming product

l_{new}^* : predicted label of the incoming product

if $l_{\text{new}}^* = t$, where $t = 1, 2, \dots, T_o$

$y_{\text{new}}^* \leftarrow \text{CNN}(X_n, \theta_s, \theta_o^{(t)})$, compute the predicted label for new data.

else if $|X_{\text{new}}| < \tilde{N}$

$X_{\text{new}} \leftarrow x_{\text{new}}$, add the incoming data to the training dataset of the new task

$t' \leftarrow \argmin_t \left\{ (1/N_t) \sum_{k=1}^{N_t} \|v_{\text{new}} - v_{k_i}^{(t)}\|_2, t = 1, 2, \dots, T \right\}$, compute the most relevant task.

$y_{\text{new}}^* \leftarrow \text{CNN}(X_n, \theta_s, \theta_o^{(t')})$, compute the predicted label for new data.

$Y_{\text{new}} \leftarrow y_{\text{new}}^*$, add the predicted label to the output of the training dataset.

else

$X_n \leftarrow X_{\text{new}}$, add X_{new} to the training dataset.

$Y_{T+1} \leftarrow Y_{\text{new}}$, add Y_{new} to the training dataset.

Train

Define $\hat{Y}_o \equiv \text{CNN}(X_n, \hat{\theta}_s, \hat{\theta}_o)$, old task output

Define $\hat{Y}_{T+1} \equiv \text{CNN}(X_n, \hat{\theta}_s, \hat{\theta}_{T+1})$, new task output

$\theta_s^*, \theta_o^*, \theta_{T+1}^* \leftarrow \argmin (\lambda_0 L_{\text{old}}(Y_o, \hat{Y}_o) + L_{\text{new}}(Y_{T+1}, \hat{Y}_{T+1}) + R(\hat{\theta}_s, \hat{\theta}_o, \hat{\theta}_{T+1}))$

end**4 Case Study**

The proposed methodology is employed across two manufacturing process case studies. In Sec. 4.1, we concentrate on the workpiece's quality prediction through the analysis of acoustic emissions generated during the LIPMM process, leveraging an advanced deep CNN. Section 4.2 is devoted to investigating the product's quality prediction by examining thermal images captured during the hot stamping process. A comparative analysis of prediction outcomes derived from the proposed continual learning model and the transfer learning model is presented. Additionally, we later implement online predictions for new, unknown tasks as they arise.

4.1 Case 1: Predictive Quality for LIPMM. LIPMM is a novel multimaterial, tool-less process that removes material at the microscale by plasma-matter interaction [35–37]. Compared with conventional laser ablation, LIPMM overcomes the shortcomings in machining materials with low absorptivity or high reflectivity [36] and has been used to machine metal alloys, transparent materials, brittle ceramics, and polymers [38,39].

In the LIPMM process, the dynamic adjustment of parameters is indispensable for optimizing machining results and achieving the desired level of quality. The pursuit of optimal parameters necessitates the exploration of new designs to fine-tune the settings effectively. However, it is imperative to acknowledge the challenges posed by factors such as laser degradation, which can lead to discrepancies between the actual laser power and the intended settings, thereby impacting machining quality. Additionally, variations in metal characteristics, such as surface roughness, exert their influence on the final machining outcome. Furthermore, the presence of randomness in the liquid medium introduces variations in the machining pattern, further affecting the overall quality. Given

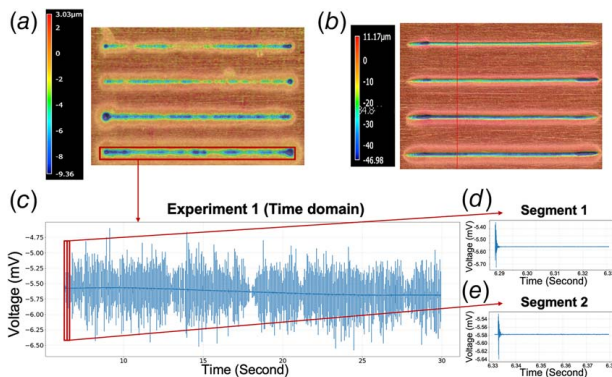


Fig. 3 Example figure demonstrating (a) bad machining; (b) good machining; (c) the raw AE signal from one of the LIPMM experiments; (d) the first segment; and (e) the second segment

the intricacies involved, the development of an automated method to detect the emergence of new machining patterns becomes essential.

This study serves as an exemplary case in which a deep learning model is leveraged to understand the relationship between process parameters and workpiece quality. The profound understanding obtained from this analysis can be readily applied to different process parameter settings, enabling the model to predict workpiece quality accurately under new and unexplored conditions.

4.1.1 LIPMM Data Description. LIPMM experiments were conducted using a Quantum Light Instruments pulsed laser (3-ns pulse duration, 50-mJ pulse energy at a wavelength of 526 nm) with a 4-mm beam diameter reduced to 60 μm at the focal point. Details on the experiment setup can be found in Ref. [40]. The workpiece was 6061 Aluminum from McMaster Carr, placed in a glass beaker on a motorized XYZ stage with 50-nm resolution, and covered with a 2-mm water layer. The optimal focal point was determined by comparing the depth of machined features at different lens distances. Microchannels were machined by moving the laser at a specified laser speed and pulse energy along the desired direction in a single pass. The dimensions of the machined channels and holes were measured using an optical profilometer. An omnidirectional hydrophone (Aquarian hydrophones, model AS-1) with a frequency range of 1 Hz to 100 kHz and a sensitivity of 40 $\mu\text{V}/\text{Pa}$ was positioned at a fixed distance from the focal spot during machining, and it was connected to a LabView device to record the acoustic signals during LIPMM. The quality of the workpieces was assessed qualitatively, with good/bad labels assigned based on the observed morphology of the machined channels. Specifically, criteria for labeling a part as “bad” included the presence of a significant heat-affected zone, thermal damage, lack of machining uniformity, and variations in depth, width, or wall smoothness. Additionally, structural weaknesses, such as cracks, were indicators of inferior quality.

An example of the machining workpiece is shown in Fig. 3(a). These samples were labeled as bad due to the nonuniformity of the surface. Other parameter combinations showed observably uniform depth along the microchannel’s length, and they were thus labeled as good (Fig. 3(b)). Figure 3(c) shows the raw AE signal from one of the experiments. The x-axis is time, and the y-axis is voltage, which is linearly proportional to the actual acoustic pressure via a known constant multiplication factor. Zooming in shows the voltage in a waveform-constant cycle. The waveform is usually between -6.5 and -4.75 mV, and then, the voltage stays stable around -5.5 mV for approximately 0.05 s. This waveform-constant cycle repeats, resulting in 828,616 data points in total for one LIPMM experiment.

To reduce the training load and increase the sample size for effective model training, we segment the AE data series into equal-length, nonoverlapping portions based on the laser pulse period. This approach not only aids in quality prediction for each segment, facilitating the identification of problematic segments for further analysis but also maintains similar waveform-constant patterns across segments, as demonstrated in Figs. 3(d) and 3(e).

Several examples of the AE segments are shown in Fig. 4. Figures 4(a) and 4(b) compare the segments from two bad-quality experiments under two different LIPMM process parameter settings, denoted as para-1 and para-2, respectively. Although they have a similar waveform-constant pattern, the waveform in Fig. 4(b) is visibly shorter and the voltages differ a lot. The voltage range in para-1 is about -5.7 mV to -5.4 mV, while the voltage range in para-2 is about -3.9 mV to -3.3 mV. Figure 4(c) shows the segment from a good-quality experiment under para-1. The voltage range is similar to that in Fig. 4(a), also para-1, which is about -5.75 mV to -5.73 mV, but the waveform pattern appears to be significantly different from the AE segments from bad-quality experiments. These three figures show that there is a relationship between AE segments and the quality of machining in LIPMM, and thus, AE segments can be used as the input in a quality prediction model. Table 2 describes the data

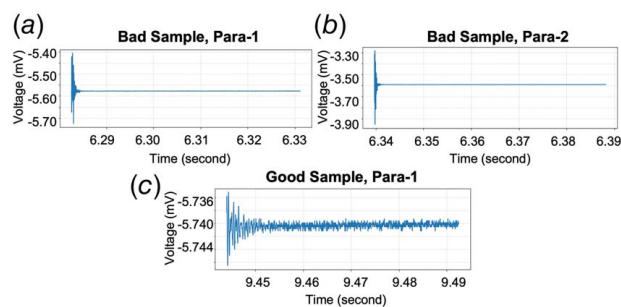


Fig. 4 AE segments from different process parameters and quality: (a) segment 1 from a bad-quality experiment under para-1; (b) segment 1 from a bad-quality experiment under para-2; and (c) segment 2 from a good-quality experiment under para-1

from each experiment, including the parameter setting index, pulse energy, laser speed, quality label, and the number of AE segments.

For the development of the base model, a substantial dataset comprising 1004 bad data and 879 good data from para-1 is utilized. Subsequently, fine-tuning of the shared layer and the acquisition of task-specific layers are performed using a more limited dataset, encompassing 102 bad data and 102 good data from both para-2 and para-3. To further augment the model’s capabilities and simulate the handling of unknown tasks, including online task prediction and quality prediction, a separate dataset containing 102 bad data and 102 good data from para-4 and para-5 each is employed.

4.1.2 Convolutional Neural Network Model for Acoustic Signal in LIPMM. CNNs are well suited for extracting informative features from complex data such as acoustic signals [41]. Various CNN models with different architectures, different input features, and different fine-tuning strategies are investigated and compared in the authors’ prior work [3], which led to the discovery that the “4Conv2h-AcouNet” architecture is most effective in learning the relationship between acoustic signals and workpiece quality. Figure 5 shows the proposed 4Conv2h-AcouNet, which consists of four convolutional layers and two hidden layers. In the proposed model, the segmented AE signals are used as input. All convolutional layers are used as the shared layers, denoted as rectangles, and all fully connected layers and hidden layers are used for task-specific layers, shown as the circles labeled with fc, h1, and h2. The output is a 2×1 vector indicating the label of micromachining quality (vector [1, 0] for bad and [0, 1] for good).

4.1.3 Case 1: Results. The performance metrics includes accuracy, precisions, negative predictive value (NPV), sensitivity, and specificity, which are calculated according to Eqs. (3)–(7). In this study, the “bad” class is positive and “good” is negative; true positive (TP) refers to the correct prediction of undesired quality, true negative (TN) refers to the correct identification of satisfying quality, and false positive (FP) and false negative (FN) are the prediction errors:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \quad (3)$$

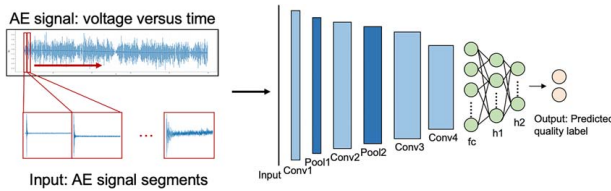
$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (4)$$

$$\text{NPV} = \frac{\text{TN}}{\text{TN} + \text{FN}} \quad (5)$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (6)$$

Table 2 Data description for quality prediction in LIPMM

Exp. #	Parameter setting (para)	Pulse energy (mJ)	Laser speed ($\times 10^{-2}$ mm/s)	Label	Number of AE segments		
					Total collected	Used for base model training	Used for continual learning
1	1	50	100	Bad	487	487	0
3	1	100	100	Bad	517	517	0
5	1	100	100	Good	422	422	0
6	1	100	100	Good	457	457	0
2	2	50	100	Bad	487	0	51
4	2	100	100	Bad	963	0	51
7	2	100	100	Good	1450	0	102
8	3	25	15	Bad	102	0	102
9	3	25	5	Good	612	0	102
10	4	40	15	Bad	102	0	102
11	4	40	5	Good	612	0	102
12	5	55	15	Bad	102	0	102
13	5	55	5	Good	612	0	102

**Fig. 5 Structure of the proposed 4Conv2h-AcouNet model**

$$\text{Specificity} = \frac{\text{TN}}{\text{FP} + \text{TN}} \quad (7)$$

The dataset is split into 80% for training and 20% for testing, where each different process parameter setting corresponds to a distinct task. After the model training phase, all testing data are treated as incoming data. Utilizing the proposed online task prediction method, each segment is assigned to the appropriate task based on the task relevance. Thereafter, the quality is predicted by employing the shared layers in conjunction with the corresponding task-specific layers.

Figure 6 illustrates the training procedure of the proposed continual learning model. The three scenarios shown in Fig. 6 are sequential, with each scenario building upon the previous one by adding the task-specific layers for a new task. In scenario I, shown in Fig. 6(a), para-1 data are used to train the model's shared and task-specific layers. Figure 6(b) illustrates scenario II, where para-2 data are added to further fine-tune the shared layers and develop task-specific layers for task 2. Similarly, Fig. 6(c) details scenario III, where para-3 data are incorporated to refine the shared layers and establish task-specific layers for task 3. All test data from para-1, para-2, and para-3 were 100% correctly classified as task 1, task 2, and task 3, respectively. The results for the three scenarios are detailed in Table 3.

Table 3(I) showcases the results of the base model trained solely on para-1 data, attaining a prediction accuracy of 96.82%. The training process for this base model requires 35.22 h. Upon introducing para-2 data for fine-tuning the shared layers and acquiring its unique task-specific layers, task 2 emerges. Table 3(II) records the model's performance in this scenario, demonstrating a remarkable prediction accuracy of 95.00%. The fine-tuning process for the shared layers and the new task-specific layer takes 12.84 h. Similarly, Table 3(III) illustrates the inclusion of para-3 data to fine-tune the shared layers and obtains its task-specific layers, resulting in the emergence of task 3. The model achieves a prediction accuracy of 97.50%, and the fine-tuning process consumes 19.26 h. As the

number of task-specific layers increases from 1 to 3, task 1 has a slight decrease in precision and specificity from 94.45% to 90.00% and from 99.07% to 90.38%, respectively, while NPV and sensitivity both rise to 100%. Similarly, for task 2, increasing from 2 to 3 task-specific layers results in a slight decrease in precision and specificity from 95.00% to 86.90% and from 95.00% to 85.00%, respectively. Nonetheless, NPV and sensitivity also increase to 100%, indicating the successful identification of all "bad" segments.

It is noteworthy that with the addition of new tasks, the prediction accuracy for task 1 exhibits a slight decline from 96.82% to 95.41% and further to 94.84%. Similarly, the prediction accuracy for task 2 experiences a marginal drop from 95.00% to 92.50%. These deviations from the initial accuracy levels occur because the addition of new tasks necessitates updating parameters in the shared layers, potentially leading them away from the previous optimal solution, θ_s^* . This phenomenon highlights the dynamic adaptability of the proposed continual learning model to accommodate novel tasks while acknowledging the slight trade-off in performance for previously established tasks.

The experimental results for tasks 1, 2, and 3 demonstrate the prediction performance of the proposed continual learning model for seen tasks. Furthermore, we demonstrate how the proposed online prediction method works for unseen tasks. Figure 7 displays the online task prediction outcomes for para-4 data, wherein the relevance thresholds for task 1, task 2, and task 3 are represented by red dash lines, specifically $\tilde{D}^{(1)} = 1.25$, $\tilde{D}^{(2)} = 8.26$, $\tilde{D}^{(3)} = 0.45$, respectively. The x-axis represents the index of the incoming data, and the y-axis represents the subspace distance between incoming data and the specified task. The points above the dashed line represent data with relevance exceeding the threshold, indicating that this data is not aligned with the specified task. Conversely, the points below the dashed line represent data with relevance within the threshold, indicating high alignment with the given task. Data points with true labels of "good" are shown on the left half of each figure (indices 1~102), while those with true labels of "bad" are shown on the right half (indices 103~204). Figures 7(a) and 7(b) reveal that the incoming segment is not assigned to either task 1 or task 2. However, in Fig. 7(c), a notable level of relevance with task 3 is observed, indicating that the incoming segments exhibit patterns similar to those of task 3. This behavior may be due to laser degradation, causing the pulse energy to drop from the intended setting of 40 mJ. Consequently, 146 new incoming segments are predicted to belong to task 3, while 58 are anticipated to belong to a previously unseen task, suggesting the emergence of a novel task.

Figure 8 illustrates the online task prediction outcomes for para-5 data. Figures 8(a) and 8(b) indicate that the new incoming segment is not assigned to either task 1 or task 2. In contrast, in Fig. 8(c),

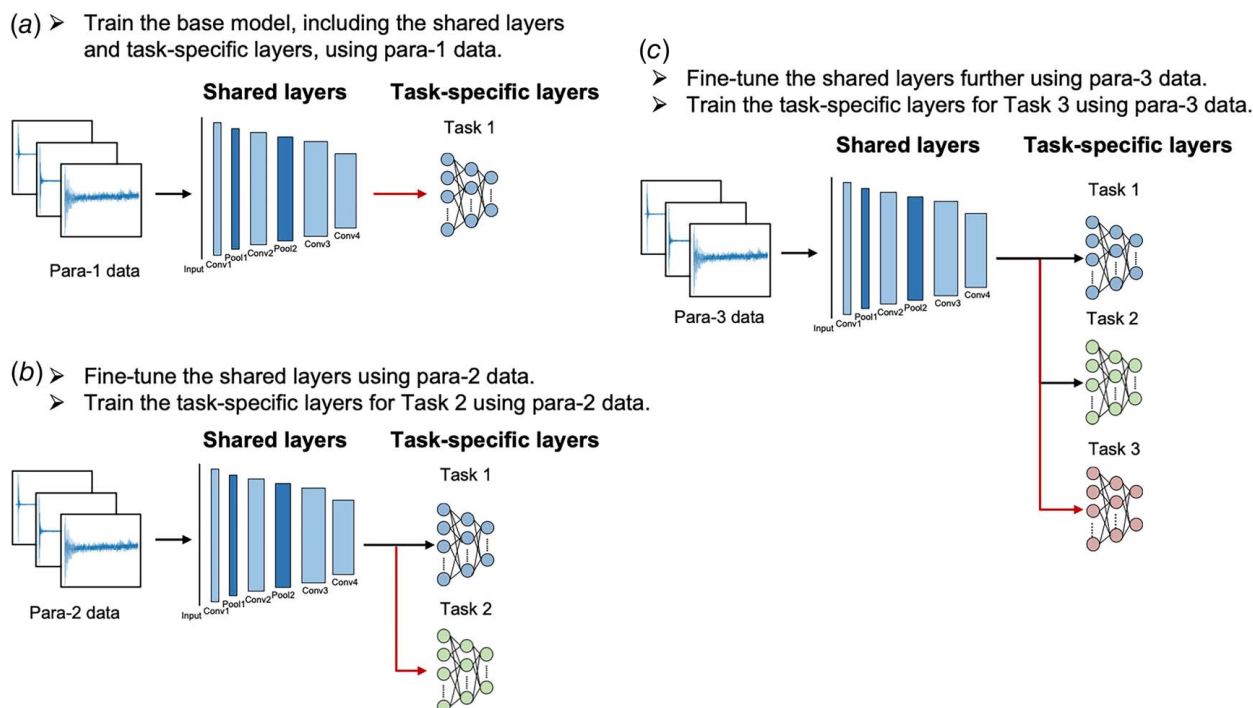


Fig. 6 Training procedure of the proposed continual learning model: (a) scenario I, (b) scenario II, and (c) scenario III

some data points fall within the relevance threshold of task 3, while Fig. 8(d) demonstrates that all new incoming data points are highly relevant to task 4. Following the proposed task prediction criteria, data points that fall below the relevance threshold of both task 3 and task 4 are assigned to the most relevant task, represented by the least subspace distance. Consequently, 204 incoming segments are assigned to task 4 as they have the least subspace distance with task 4. This observation echoes the previous analysis, suggesting that laser degradation during machining may be influencing the energy levels, resulting in a deviation from the intended setting of 55 mJ and contributing to the observed pattern. The online task prediction allows for the dynamic allocation of new data points to their most appropriate tasks based on their relevance, enabling the model to adeptly handle the emergence of novel tasks. Meanwhile, it enhances the model's robustness in accommodating variations in the machining process and the influence of laser degradation on the quality of the machined components.

Figure 9 demonstrates the quality prediction process employed by our proposed online continual learning model. In scenario IV, as depicted in Fig. 9(a), 146 incoming AE segments are assigned to task 3, and the predictions are made using the task 3-specific layers. Figure 9(b) showcases scenario V, where the initial 50 incoming AE segments are identified as belonging to task 4 but are predicted using the task 3-specific layers due to their high relevance to this task. Figure 9(c) continues in scenario V, with the remaining 212 incoming AE segments also designated to task 4,

but in this instance, predictions are made using the newly trained task 4-specific layers. The results from these scenarios are summarized in Table 4.

Table 4(IV) shows the results for scenario IV. The quality prediction for these data points is conducted using the shared layers and task-specific layers tailored for task 3, achieving an accuracy of 95.21%. Given the emergence of task 4, the challenge lies in the insufficient training data available for the task-specific layers of this new task. To address this, the first $\tilde{N}$ data points of task 4 are predicted using the most relevant task, which in this case is task 3. The threshold $\tilde{N}$ was determined based on the production scenario and considerations of model training efficiency. Given the assumption that each production batch consists of 50 units, setting $\tilde{N} = 50$ is a practical choice because it aligns with the natural data accumulation cycle in the production process. This allows the model to be updated with each batch, ensuring timely training without unnecessary delays. In addition, the threshold was carefully selected to balance the need for sufficient data to optimize task-specific parameters with practical constraints, such as time and computational resources. A smaller dataset may lack the necessary information for effective model training, while a larger dataset would require more time to accumulate and increase computational costs. Thus, $\tilde{N} = 50$ strikes a reasonable balance between model performance and computational efficiency.

As shown in Table 4(V), the prediction performance for the first 50 data points of task 4, utilizing the task 3 model, achieves an

Table 3 Prediction performance of the proposed continual learning model with para-1, para-2, and para-3 data

Scenario	Task-specific layers trained by	Accuracy	Precision	NPV	Sensitivity	Specificity	Training time (h)
Scenario I	Task 1	0.9682	0.9445	0.9941	0.9180	0.9907	35.22
Scenario II	Task 1	0.9541	0.9050	0.9893	0.9180	0.9907	12.84
	Task 2	0.9500	0.9500	0.9500	0.9500	0.9500	
Scenario III	Task 1	0.9484	0.9000	1.0000	1.0000	0.9038	19.26
	Task 2	0.9250	0.8690	1.0000	1.0000	0.8500	
	Task 3	0.9750	0.9520	1.0000	1.0000	0.9500	

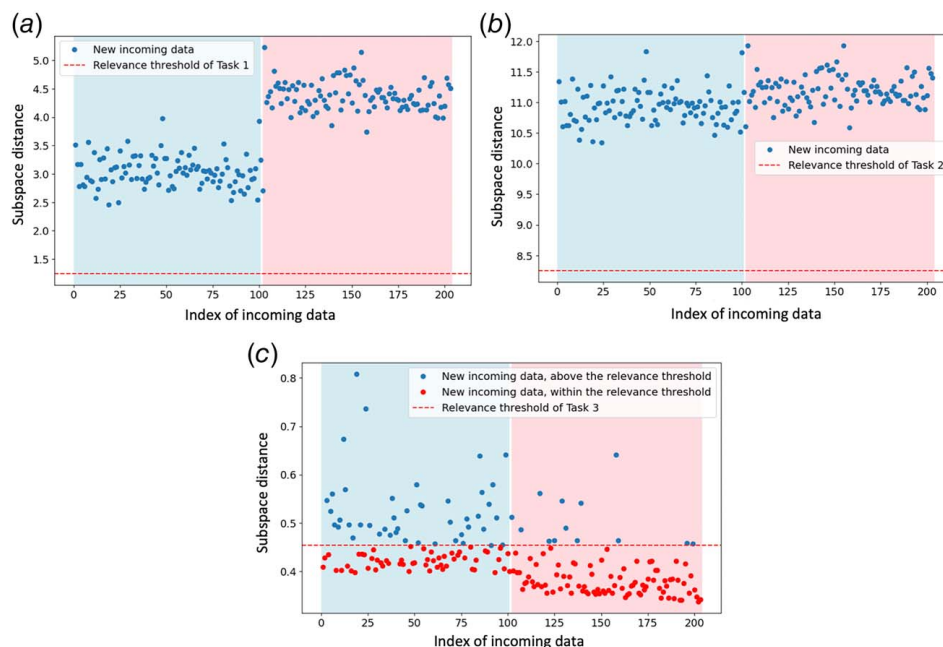


Fig. 7 Task prediction process for incoming segments (para-4 data). (a) Task relevance with task 1 versus relevance threshold of task 1. (b) Task relevance with task 2 versus relevance threshold of task 2. (c) Task relevance with task 3 versus relevance threshold of task 3.

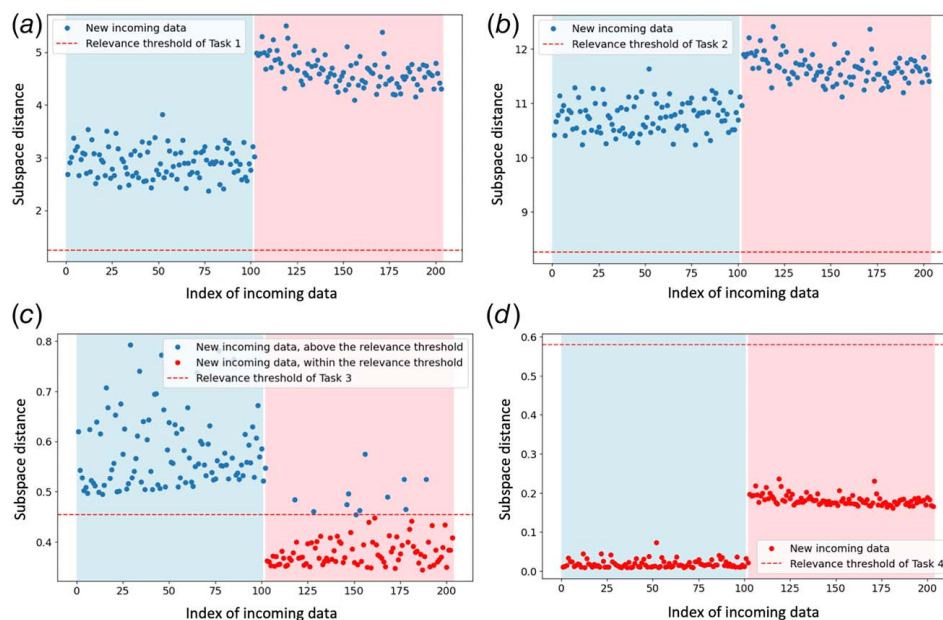


Fig. 8 Task prediction process for incoming segments (para-5 data). (a) Task relevance with task 1 versus relevance threshold of task 1. (b) Task relevance with task 2 versus relevance threshold of task 2. (c) Task relevance with task 3 versus relevance threshold of task 3. (d) Task relevance with task 4 versus relevance threshold of task 4.

accuracy of 89.93%. Subsequently, the 50 data points are utilized to obtain the task-specific layer of task 4, and the remaining data points of task 4 are then predicted using the task 4 model. Table 4(VI) showcases the prediction accuracy for task 4. Among the 212 segments, 8 are from para 4 and 204 are from para-5. The accuracy slightly drops from 89.93% to 86.00% with the precision and specificity dropping to 78.12% and 72.00%, respectively, indicating an increasing number of misclassified “good” segments as the number of task-specific layers increases, while NPV and sensitivity still keep 100%, meaning that no “bad” segments are misclassified as “good.” Overall, the model demonstrates the effectiveness of detecting “bad” segments.

We compare the proposed method with the transfer learning method in Table 5. Table 5(a) indicates the prediction performance with the transfer learning method. In the transfer learning approach, para-1 data are used to train the base model, and predictions for other tasks are achieved by fine-tuning the fully connected layer and hidden layers using para-2 and para-3 data, respectively. Notably, there is no separate model for each task; instead, the same model is incrementally updated as new tasks are introduced. Additionally, since the transfer learning method cannot identify unseen tasks, predictions for task 4 are based on the model trained for task 3. In terms of the training time, transfer learning has less time cost. For example, it takes 7.72 h to transfer from

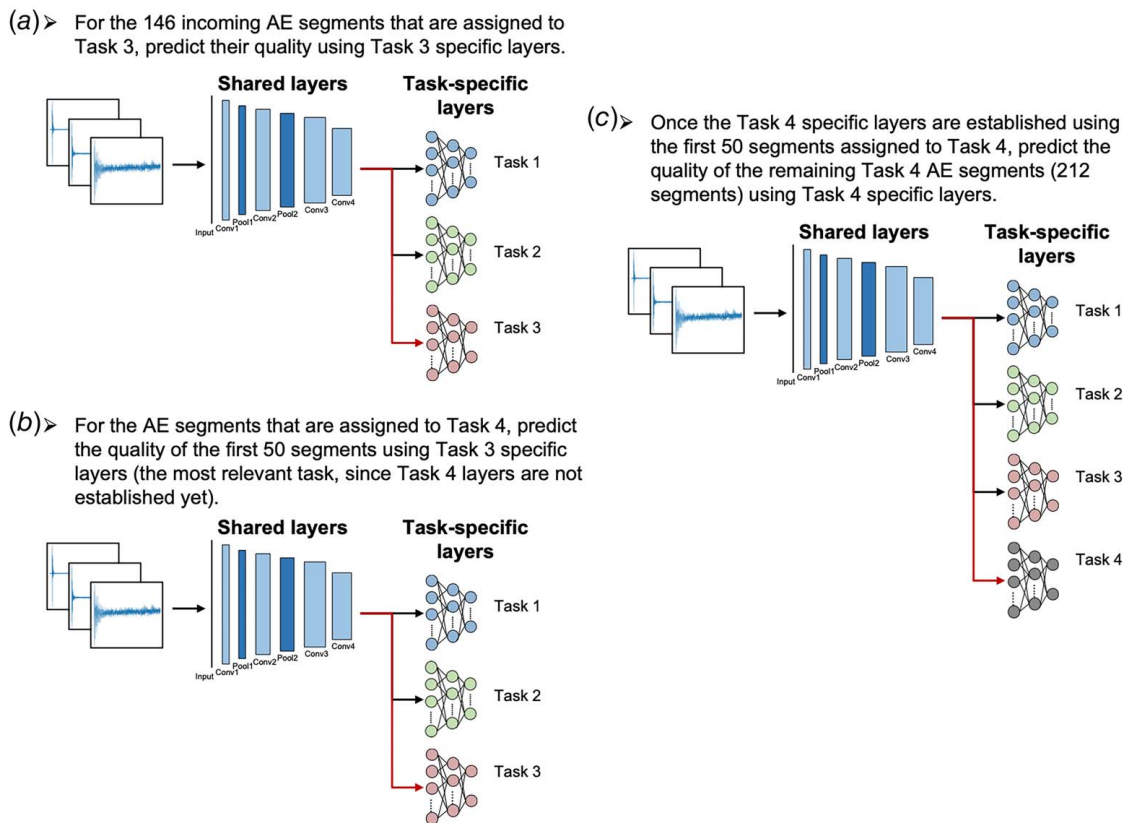


Fig. 9 Prediction procedure of the proposed online continual learning model: (a) scenario IV, (b) scenario V, and (c) scenario VI

the task 1 model to the task 2 model in transfer learning, while it takes 12.84 h to train the continual learning model. This discrepancy arises because the transfer learning model only fine-tunes the last few layers, whereas the continual learning model necessitates updates to the shared layers and fine-tuning of the task-specific layers, resulting in a greater number of parameters to optimize.

Despite the shorter training time, the transfer learning method exhibits issues related to catastrophic forgetting. When the model is transferred to task 3, the prediction performance for task 1 and task 2 experiences a decline, reaching 86.27% and 81.37% accuracy, respectively. In contrast, the proposed continual learning model effectively avoids such forgetting, maintaining prediction accuracies of 94.84% and 96.13% for task 1 and task 2, respectively. Furthermore, the proposed method effectively handles newly emerged unseen tasks, achieving an 86% prediction accuracy for task 4. In comparison, the transfer learning method achieves only 78.43% prediction accuracy for task 4. This observation demonstrates the effectiveness of the proposed continual learning model in adapting to novel tasks and its ability to achieve accurate predictions.

4.2 Case 2: Predictive Quality for Hot Stamping Process.

Hot stamping is a forming technique of a thinner flat part that consists of the approach phase, part forming and part quenching [11]. It is widely used in automotive manufacturing [42]. In the context of

hot stamping processes, the emergence of new target products or variations in machine conditions during production poses a challenge, rendering previously trained models unsuitable for predicting the quality of these new tasks. The tools and machines utilized in hot stamping may undergo periodic or irregular renewals, leading to the reemergence of tasks with the same product or process design. Due to the significant time and labor costs associated with monitoring the production status of each individual product, unknown tasks often arise during the manufacturing process, making it difficult to develop a suitable predictive model. This case study is used to demonstrate how the proposed online prediction algorithm is implemented to address this issue. In this case, the proposed CNN model is used to understand the relationship between operational conditions of machine under variations of target products through hot stamping thermal images. Each target product possesses its own unique set of potential problems, and this distinctiveness is effectively captured and learned by the task-specific layers within the CNN model. By integrating this online prediction algorithm with the proposed CNN model, our case study showcases its prowess in effectively predicting the quality outcomes for new target products and variations in machine conditions during the hot stamping process.

4.2.1 Hot Stamping Thermal Image Data Description. The hot stamping process involves heating a metal blank to a temperature range of 900–950 °C over a period of 5–10 min in a furnace until

Table 4 Prediction performance of the proposed online continual learning model with para-4 and para-5 data

Implementation	Task-specific layers trained by	Accuracy	Precision	NPV	Sensitivity	Specificity
Scenario IV	Task 3	0.9521	0.9041	1.0000	1.0000	0.9125
Scenario V	Task 3	0.9000	0.8333	1.0000	1.0000	0.8000
Scenario VI	Task 4	0.8600	0.7812	1.0000	1.0000	0.7200

full austenitization is achieved. Once heated, the hot sheet is rapidly transferred from the furnace to the press, where it is immediately stamped. The stamping tool remains closed for approximately 20 s to quench the sheet below the martensite finish temperature. To collect real-time production data, an in-process pyrometer is used to capture the thermal image of each stamped part. The obtained data come in a matrix form whose elements are the temperature of the part surface. A visual comparison in Fig. 10 shows that products in different tasks differ in both surface temperature and shape. The x -axis and y -axis indicate the index of pixels of the thermal image. The raw thermal image of a task 1 product is 650 pixels $\times$ 290 pixels, while that for task 2, 3, and 4 products is 700 pixels $\times$ 250 pixels, 700 pixels $\times$ 250 pixels, and 798 pixels $\times$ 399 pixels, respectively. Throughout the hot stamping process, several factors can influence the final quality of the products. These include overheating, tool wear, abnormal heating time, and holding time, all of which can result in undesired outcomes. Expert guidance with domain knowledge is employed to assign quality labels to the thermal images. This ensures that the quality assessment is informed by a rich understanding of the hot stamping process, thus enabling accurate and reliable quality classification.

Table 6 illustrates the data employed to train and test the proposed continual learning model. A noteworthy imbalance between “good” and “bad” data is observed in the raw dataset. To address

the issue of insufficient data and achieve a more balanced data distribution, we employed bootstrapping as our data augmentation technique. Bootstrapping, which involves resampling with replacement from the existing data, effectively increases the dataset size while maintaining the original distribution. This method allowed us to enhance the quantity of training data to meet the predetermined levels required for robust model training. Specifically, the raw “bad” data for task 1 were augmented from 400 to 500, for task 2 from 29 to 100, and for task 3 from 7 to 85. Task 1 data are used for obtaining the base model, while task 2 and task 3 data are used for continual learning for seen tasks. Finally, task 4 data are used to mimic the online prediction for a newly emerged task.

4.2.2 Convolutional Neural Network Model for Hot Stamping Thermal Images. The architecture of the proposed CNN model named ThermNet for quality prediction is shown in Fig. 11. The input of the CNN model is a 128 $\times$ 128 three-channel hot stamping thermal image. The image is projected and reshaped through five pairs of convolutional layers and max-pooling layers, followed by two fully connected layers with 2048 channels each. All convolutional layers and fully connected layers have a ReLU activation function. Meanwhile, a 1×12 vector F_i is added to the last fully connected layer, represented by Eq. (8).

$$F_i = [H_i^{(h)}, H_i^{(75\%)}, H_i^{(m)}, H_i^{(25\%)}, H_i^{(l)}, H_i^{(var)}, P_i^{(100)}, P_i^{(120)}, P_i^{(140)}, P_i^{(160)}, P_i^{(180)}, P_i^{(200)}] \quad (8)$$

F_i comprises two sets of features. The first set consists of descriptive statistics, including the maximum ($H_i^{(h)}$), 75th percentile ($H_i^{(75\%)}$), the medium ($H_i^{(m)}$), 25th percentile ($H_i^{(25\%)}$), the minimum ($H_i^{(l)}$), and the variance ($H_i^{(var)}$). The second set encompasses physics-informed features, specifically pertaining to the heat distribution on the hot stamping part, which is calculated by the area of the heat-affected zone (HAZ) above different temperature thresholds, $\tilde{T}$ (where $\tilde{T} = 100, 120, 140, 160, 180, 200$). The area of HAZ is quantified by the number of pixels on the thermal image that exhibit temperatures higher than the specified thresholds. The fused layer, derived from the combination of these features, is subsequently projected into a 1×2 vector, employing a softmax activation function. This vector effectively represents the estimated probabilities for the “bad” and “good” classes corresponding to the input image. Ultimately, the image is assigned to the class with the higher estimated probability. In training the CNN model, we used the cross-entropy loss function and the stochastic gradient descent optimizer.

4.2.3 Case 2: Results. Testing data from all four tasks are 100% correctly classified. Figure 12 displays the online task prediction outcomes for task 4 data, and the relevance thresholds for task 1, task 2, and task 3 are represented by red dash lines, $\tilde{D}^{(1)} = 6.04$, $\tilde{D}^{(2)} = 0.21$, $\tilde{D}^{(3)} = 15.77$, respectively. Figures 12(a)–

12(c) reveal that the incoming product is not assigned to any of the existing tasks, indicating the emergence of a new task. Given limited incoming products, there are no sufficient data for training corresponding task-specific layers. Therefore, the task-specific layers linked with the most relevant task, which has the least subspace distance, are utilized for quality prediction.

The comparison between the proposed method and the transfer learning method is shown in Table 7. The training time cost of transfer learning for task 2 and task 3 is 0.73 h compared with 7.74 and 9.60 h of the proposed continual learning method, but this is under the sacrifice of prediction accuracy. This is because the shared layers capture critical features, so that continual learning gets more chance to the optimal parameters. In addition, catastrophic forgetting can be observed in transfer learning. For example, the prediction accuracy of task 2 drops from 85.0% to 72.5% when the model is transferred from task 2 to task 3. However, the proposed continual learning method maintains a relatively stable prediction performance with the increment of tasks, with task 2 accuracy only slightly decreasing from 97.50% to 95.00% after transferring to task 3. Furthermore, the online prediction accuracy for task 4 achieves 83.33% with the proposed method, compared to just 62.50% with transfer learning. The results affirm the potential of the proposed continual learning approach as an effective solution for in-situ quality prediction in manufacturing processes in the face of emerging tasks.

Table 5 Performance comparison between the proposed method and transfer learning method

Task no.	(a) Prediction accuracy with transfer learning method					(b) Prediction accuracy with the proposed method				
	Prediction for each task				Training time (h)	Prediction for each task				Training time (h)
	1	2	3	4		1	2	3	4	
1	0.9682	–	–	–	35.22	0.9682	–	–	–	35.22
2	0.9264	0.9448	–	–	7.72	0.9541	0.9500	–	–	12.84
3	0.8627	0.8137	0.9390	–	7.72	0.9484	0.9613	0.9750	–	19.26
4	0.8627	0.8137	0.9390	0.7843	–	0.9264	0.9362	0.9750	0.8600	4.21

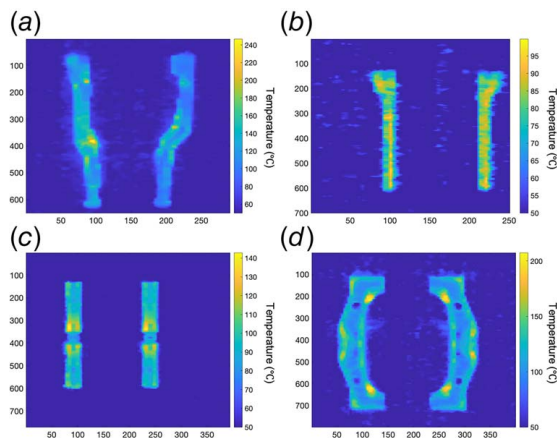


Fig. 10 Thermal images of products in different tasks: (a) task 1; (b) task 2; (c) task 3; and (d) task 4

5 Conclusions and Future Work

Modern manufacturing faces the challenge of adapting to frequent product/process changes, undetected machine degradation, and fluctuations in the machining environment. These issues reduce the reliability of established models for quality prediction and lead to new, unpredictable production patterns. Our study

Table 6 Data used for training and testing the proposed continual learning model

Type	Raw data		Used for training		Used for testing	
	Good	Bad	Good	Bad	Good	Bad
Task 1	623	490	400	400	100	100
Task 2	616	29	80	80	20	20
Task 3	93	7	80	80	5	5
Task 4	11	13	—	—	11	13

introduces a new continual learning approach that adapts to these changing conditions in manufacturing. This method includes an online algorithm to predict current production patterns and a continual learning-enhanced quality prediction model that remembers previously acquired knowledge. The proposed online continual learning approach effectively addresses new tasks as they arise, offering a dependable way to deal with manufacturing process variations. We have tested our method with two case studies: one examining machine degradation and the other exploring changes in target products. The results demonstrate the superiority of the proposed continual learning method over the transfer learning approach. By avoiding catastrophic forgetting, achieving accurate predictions for known tasks, and adeptly accommodating new and unseen tasks, the proposed continual learning model emerges as a

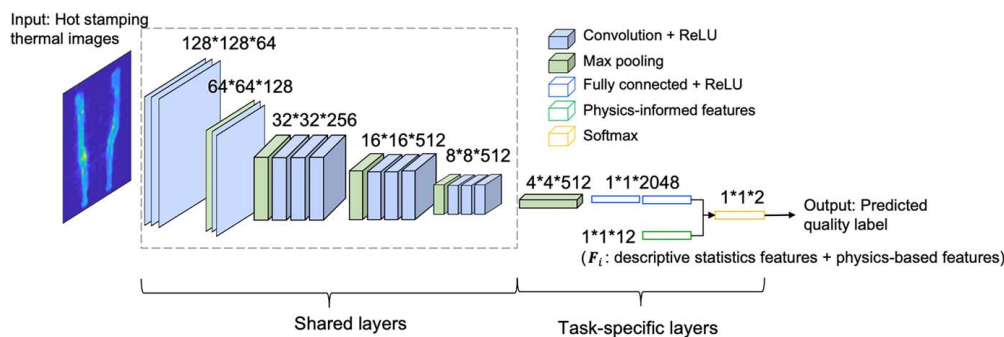


Fig. 11 Architecture of the proposed CNN model named ThermNet

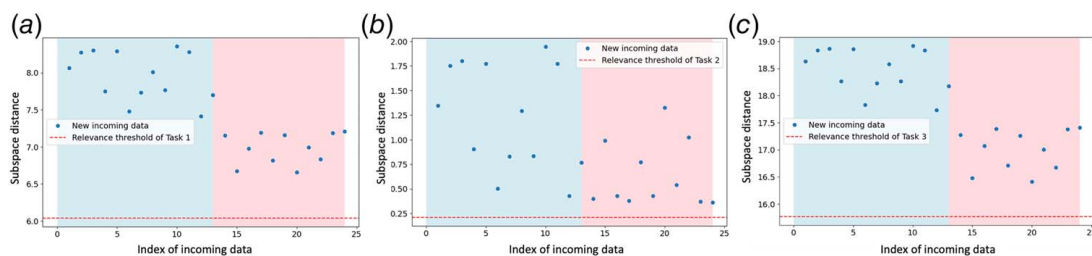


Fig. 12 Task prediction process for incoming segments (task-4 data). (a) Task relevance with task 1 versus relevance threshold of task 1. (b) Task relevance with task 2 versus relevance threshold of task 2. (c) Task relevance with task 3 versus relevance threshold of task 3.

Table 7 Performance comparison between the proposed method and transfer learning method

Task no.	(a) Prediction accuracy with transfer learning method					(b) Prediction accuracy with the proposed method				
	Prediction for each task				Training time (h)	Prediction for each task				Training time (h)
	1	2	3	4		1	2	3	4	
1	0.9150	—	—	—	19.34	0.9150	—	—	—	19.34
2	0.9150	0.8500	—	—	0.73	0.9150	0.9750	—	—	7.74
3	0.9150	0.7250	0.8250	—	0.73	0.8950	0.9500	0.9000	—	9.60
4	0.9150	0.7250	0.8250	0.6250	—	0.8950	0.9500	0.9000	0.8333	—

powerful and reliable tool for in-situ quality prediction in manufacturing processes.

The proposed method is applicable in discrete manufacturing environments equipped with sensors and requiring quality prediction. The method is particularly well-suited and effective for manufacturing environments that need to adapt to variations, such as parameter changes or the emergence of new tasks. The approach can be applied in various industrial settings, including forming and machining. For instance, in automotive stamping, where the production line may need to produce different shapes of parts at different times (e.g., customized roof designs), the method allows for immediate quality assessment of these parts, eliminating lengthy model training periods. This capability supports continuous upgrades and production of new designs with efficient monitoring. The method also adapts to changes in production patterns due to tool degradation or replacement. However, it relies on high-quality sensors for accurate data collection and should be tailored to the system's characteristics. For simpler processes with infrequent design changes or no reemergent tasks, transfer learning might be more suitable due to its shorter training time.

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

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